

IT 326: Data Mining – Final Report

Alzheimer's Disease

Semester: 1st Semester 1447 H

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1. Problem

Alzheimer's Disease remains a major public health challenge, affecting millions worldwide and placing emotional and financial strain on families and healthcare systems. Its early symptoms are subtle, making timely diagnosis difficult.

This project aims to use data mining techniques to explore patient characteristics and develop predictive models that determine whether an individual is likely to have Alzheimer's.

Understanding these patterns is crucial, as early detection leads to more effective treatment planning, improved quality of life, and better long-term support for patients and caregivers.

2. Data Mining Task

This project focuses on two main data mining techniques: classification and clustering, both used to analyze and understand Alzheimer's Disease patterns.

For classification, a Decision Tree model was developed to determine whether a patient is diagnosed with Alzheimer's. The prediction relies on several medical, lifestyle, and cognitive features —such as age, family history, cholesterol levels, blood pressure, sleep quality, and MMSE scores. The target variable for this task is Diagnosis, which identifies each individual as either having Alzheimer's (1) or not (0).

For clustering, we applied the K-Means algorithm to group patients based on similarities in their attributes, but without using the diagnosis label. This unsupervised approach helps reveal natural groupings in the data and highlights shared characteristics among patients. These clusters can provide useful insights into potential risk factors and may help identify patterns that contribute to early detection and preventive strategies.

3. Data

- **Source:** [Kaggle Alzheimer's Dataset](#)
- **Number of features :** 35
- **Number of records :** 2149
- **Class label: Diagnosis (0 = No Alzheimer's, 1 = Alzheimer's)**
- **Other features:** include age, gender, BMI, blood pressure, cholesterol, sleep quality, physical activity, and family history of Alzheimer's.

Diagnosis	
0	1389
1	760

Attributes with their data types:

NameOfAttribute	DataType
0 PatientID	int64
1 Age	int64
2 Gender	int64
3 Ethnicity	int64
4 EducationLevel	int64
5 BMI	float64
6 Smoking	int64
7 AlcoholConsumption	float64
8 PhysicalActivity	float64
9 DietQuality	float64
10 SleepQuality	float64
11 FamilyHistoryAlzheimers	int64
12 CardiovascularDisease	int64
13 Diabetes	int64
14 Depression	int64
15 HeadInjury	int64
16 Hypertension	int64
17 SystolicBP	int64
18 DiastolicBP	int64
19 CholesterolTotal	float64
20 CholesterolLDL	float64
21 CholesterolHDL	float64
22 CholesterolTriglycerides	float64
23 MMSE	float64
24 FunctionalAssessment	float64
25 MemoryComplaints	int64
26 BehavioralProblems	int64
27 ADL	float64
28 Confusion	int64
29 Disorientation	int64
30 PersonalityChanges	int64
31 DifficultyCompletingTasks	int64
32 Forgetfulness	int64
33 Diagnosis	int64
34 DoctorInCharge	object

Number of attributes: 35

Statistical Measures

We computed descriptive statistics to understand data distribution.

Five-number summary for selected numeric features							
	Age	BMI	AlcoholConsumption	PhysicalActivity	DietQuality	SleepQuality	FunctionalAssessment
min	60	15	0	0	0	4	0
25%	67	21	5.13	2.57	2.45	5.48	2.5
50%	75	27	9.93	4.76	5.07	7.11	5.09
75%	83	33.86	15.15	7.42	7.55	8.56	7.54
max	90	39.99	19.98	9.98	9.99	9.99	9.99
	SystolicBP	DiastolicBP	CholesterolTotal	CholesterolLDL	CholesterolHDL	MMSE	ADL
min	90	90	150.09	50.23	20	0	0
25%	112	74	190.25	87.19	39.09	7.16	2.3
50%	134	91	225.08	123.34	59.76	14.4	5.03
75%	157	105	262.03	161.73	78.93	22.16	7.58
max	179	119	299.99	199.96	99.98	29.99	9.99

Mean and Standard Deviation

To further describe the dataset, we calculated the **mean** and **standard deviation** for each numeric attribute.

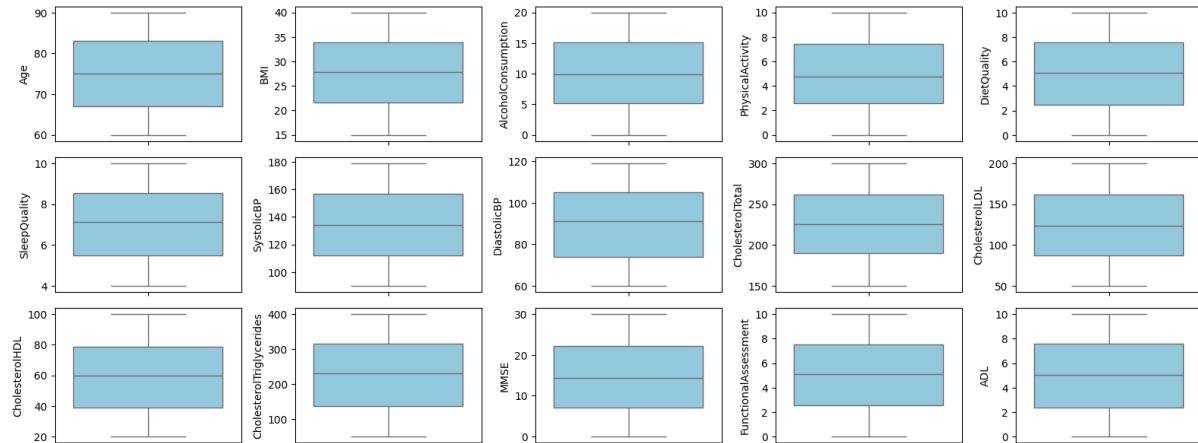
	Mean	Standard Deviation
PatientID	5825.000000	620.507185
Age	74.908795	8.990221
Gender	0.506282	0.500077
Ethnicity	0.697534	0.996128
EducationLevel	1.286645	0.904527
BMI	27.655697	7.217438
Smoking	0.288506	0.453173
AlcoholConsumption	10.039442	5.757910
PhysicalActivity	4.920202	2.857191
DietQuality	4.993138	2.909055

Outliers

Outlier detection using **Z-score** method and **IQR** method

Results: No outliers detected in any numeric column.

This indicates that the dataset is well-distributed and ready for modeling without removal or correction.



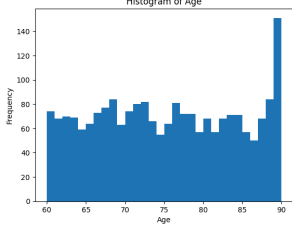
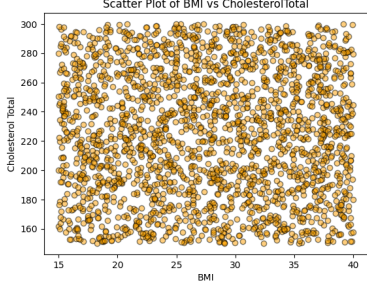
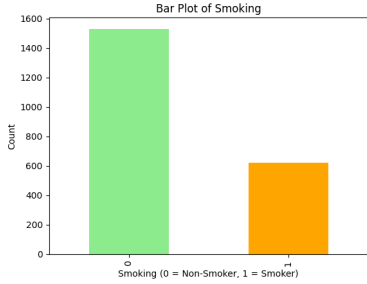
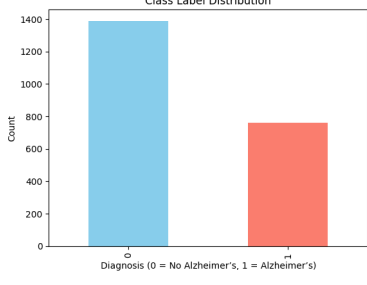
Missing Values

The dataset contains no missing values.

All columns were complete in the raw dataset.

CholesterolLDL	0	PatientID	0
CholesterolHDL	0	Age	0
CholesterolTriglycerides	0	Gender	0
MMSE	0	Ethnicity	0
FunctionalAssessment	0	EducationLevel	0
MemoryComplaints	0	BMI	0
BehavioralProblems	0	Smoking	0
ADL	0	AlcoholConsumption	0
Confusion	0	PhysicalActivity	0
Disorientation	0	DietQuality	0
PersonalityChanges	0	SleepQuality	0
DifficultyCompletingTasks	0	FamilyHistoryAlzheimers	0
Forgetfulness	0	CardiovascularDisease	0
Diagnosis	0	Diabetes	0
DoctorInCharge	0	Depression	0
		HeadInjury	0
		Hypertension	0
		SystolicBP	0
		DiastolicBP	0
		CholesterolTotal	0
		CholesterolLDL	0

Visual Data Representation

Plot	Image	Description
Histogram		The histogram shows most patients are between 60–90 years old, with many clustered near age 90. This confirms the dataset is concentrated on older adults and supports using normalization or discretization.
Scatter Plot		The scatter plot indicates a positive relationship: higher BMI is often associated with higher cholesterol. Since both are meaningful risk factors, normalization or outlier handling may be necessary.
Bar Plot		The plot shows far more non-smokers than smokers. This imbalance suggests the feature may require careful preprocessing to avoid bias in analysis.
Class Label Distribution (Bar Plot)		The bar plot shows that class 0 (No Alzheimer's) appears more frequently than class 1 (Alzheimer's), indicating class imbalance. This highlights the need for preprocessing steps to support accurate modeling.

4. Data Preprocessing

The following preprocessing steps were applied:

1. Discretization

We converted continuous Age into an AgeGroup feature using fixed ranges (<60, 60–69, 70–79, 80+). This makes the feature more interpretable and helps highlight clearer age-related patterns for analysis.

	Age	AgeGroup
1051	90	80+
1653	63	60–69
400	74	70–79
682	62	60–69
1046	75	70–79

2. Normalization

We applied Min–Max normalization to continuous numerical features only, scaling them to [0, 1]. Categorical fields and identifiers (e.g PatientID, AgeGroup, Gender) were excluded. This prevents large-range features like cholesterol and blood pressure from dominating the model.

	BMI	AlcoholConsumption	PhysicalActivity	DietQuality	SleepQuality	SystolicBP	DiastolicBP	CholesterolTotal	Chole
369	0.670005	0.416565	0.357627	0.037540	0.670564	0.179775	0.355932	0.579202	
784	0.888409	0.793737	0.083221	0.740723	0.818305	0.438202	0.881356	0.459010	
69	0.354746	0.307429	0.387976	0.063098	0.343539	0.797753	0.508475	0.225612	
1384	0.637248	0.954526	0.734716	0.391968	0.588336	0.168539	0.542373	0.619911	
1836	0.106037	0.480523	0.863591	0.261098	0.022465	0.449438	0.152542	0.660881	

3. Feature Selection

Categorical features were encoded using one-hot encoding, and the target Diagnosis was converted to numeric codes. We then applied SelectKBest (ANOVA F-test) to keep the top 20 most relevant features. This reduced dimensionality and retained only the strongest predictors.

Out[21]:	PatientID	Gender	EducationLevel	BMI	SleepQuality	FamilyHistoryAlzheimers	CardiovascularDisease	Diabetes	HeadInjury
	913	5664	0	1	0.539664	0.696222	0	0	0
	901	5652	1	2	0.890785	0.645624	1	0	0
	122	4873	1	1	0.816918	0.477215	0	0	0
	603	5354	1	2	0.461321	0.965622	0	0	0
	1634	6385	1	0	0.797508	0.386378	0	0	0

5 rows × 21 columns

Data before preprocessing

Snapshot of Raw Dataset:											
	PatientID	Age	Gender	Ethnicity	EducationLevel	BMI	Smoking	AlcoholConsumption	PhysicalActivity	DietQuality	...
0	4751	73	0	0	2	22.927749	0	13.297218	6.327112	1.347214	...
1	4752	89	0	0	0	26.827681	0	4.542524	7.619885	0.518767	...
2	4753	73	0	3	1	17.795882	0	19.555085	7.844988	1.826335	...
3	4754	74	1	0	1	33.800817	1	12.209266	8.428001	7.435604	...
4	4755	89	0	0	0	20.716974	0	18.454356	6.310461	0.795498	...

5 rows × 35 columns

Data after being preprocessed

Snapshot of Preprocessed Dataset:										
	PatientID	Gender	EducationLevel	BMI	SleepQuality	FamilyHistoryAlzheimers	CardiovascularDisease	Diabetes	HeadInjury	
0	4751	0	2	0.316960	0.837564	0	0	1		
1	4752	0	0	0.473058	0.525021	0	0	0		
2	4753	0	1	0.111553	0.945597	1	0	0		
3	4754	1	1	0.752163	0.731994	0	0	0		
4	4755	0	0	0.228472	0.265892	0	0	0		

5 rows × 21 columns

5. Data Mining Techniques

We applied two data mining techniques in this project: **Decision tree** for **Classification** (Supervised Learning) and **K-Means** for **Clustering** (Unsupervised Learning). Both techniques were implemented using Python packages such as scikit-learn, pandas, numpy, and matplotlib. Below is a summary of the packages and methods used.

All packages and methods used:

Package	Methods Used
scikit-learn (sklearn)	<code>train_test_split()</code> , <code>DecisionTreeClassifier()</code> , <code>accuracy_score()</code> , <code>confusion_matrix()</code> , <code>plot_tree()</code> , <code>MinMaxScaler()</code> , <code>silhouette_score()</code> , <code>silhouette_samples()</code> , <code>KMeans()</code> , <code>PCA()</code>
pandas	Used for loading and manipulating the dataset (<code>read_csv()</code> , <code>DataFrame()</code> , inspecting data types, handling columns).
numpy	Provides numerical operations and supports arrays used internally by machine learning algorithms.
matplotlib	Used to create visualizations such as scatter plots, bar charts, and histograms.
seaborn	Provides advanced statistical visualizations and improves the styling of plots during data exploration.

1. Classification Technique

In this part of the project, we used the **Decision Tree classifier** to predict the Diagnosis class, where the target variable indicates whether a patient has Alzheimer's (1) or does not have Alzheimer's (0). The Decision Tree algorithm was chosen because it is easy to interpret, handles both numerical and categorical attributes, and clearly illustrates the decision-making process through its hierarchical structure. All demographic, medical, lifestyle, and cognitive features were used as input variables, while the Diagnosis column served as the output class.

To evaluate the model's performance under different data conditions, we divided the dataset into three train-test splits:

- 90% training with 10% testing
- 80% training with 20% testing
- 70% training with 30% testing

Using multiple ratios allowed us to observe how the model behaves with varying amounts of training data and to identify the most reliable configuration.

The classifier was trained using two attribute selection criteria: **Information Gain** (Entropy) and the **Gini Index**. For each combination of split ratio and criterion, the model was trained, predictions were generated, accuracy was calculated, and the confusion matrix was analyzed. Additionally, the decision tree visualization helped us understand how the model separated Alzheimer's and non-Alzheimer's cases based on the most significant features.

To compare the performance of all configurations, we relied on accuracy scores and confusion matrix results. These evaluation metrics provided a clear understanding of how effectively the Decision Tree distinguished between the two classes and helped identify the best-performing split ratio and criterion for Alzheimer's prediction.

2. Clustering Technique

Clustering is an unsupervised learning approach, meaning it does not require a target label. For this reason, the Diagnosis column was removed before applying the clustering algorithm. We also excluded the PatientID column since it is simply an identifier and does not contribute to the clustering process. After this adjustment, the dataset contained only relevant numerical features, all of which were already normalized during the preprocessing to ensure equal contribution from each attribute.

In this project, we applied the **K-Means** clustering algorithm, which groups data points into K clusters based on their similarity. The algorithm begins by assigning data points to initial cluster centroids and then iteratively updates these centroids by minimizing the distance between each point and its assigned cluster. This process continues until the cluster assignments stabilize and the within-cluster variation is minimized.

To determine an appropriate number of clusters for our dataset, we evaluated three different values of K: 2, 3, and 4. For each value, we measured clustering performance using two widely adopted evaluation metrics. The first metric, the **Silhouette Score**, evaluates how well-separated and coherent the clusters are by comparing the distance between points in the same cluster versus points in other clusters. Higher scores indicate clearer and more

meaningful grouping. The second metric, **Inertia** (also known as Within-Cluster Sum of Squares), was used as part of the Elbow Method to assess how compact the clusters are and to identify the point where increasing the number of clusters no longer produces a significant improvement.

After testing the three values of K, we analyzed the results and identified which configuration achieved the best balance of separation and compactness. These cluster assignments were then examined to understand the characteristics and patterns within each group, offering insights into similarities among patients and potential Alzheimer's risk profiles.

6. Evaluation and Comparison

This section presents and compares the results obtained from both the Decision Tree and K-Means models. Each model was evaluated using appropriate metrics and visualizations to understand its performance and ability to identify Alzheimer's Disease patterns.

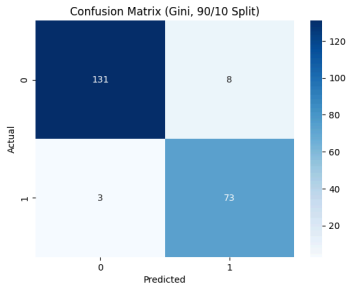
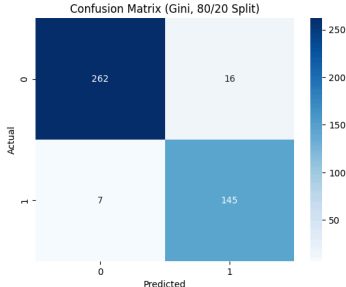
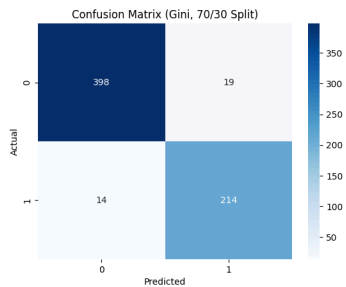
A. Classification Results - Decision Tree

The Decision Tree classifier was evaluated using three different train-test split ratios (90/10, 80/20, and 70/30) and two splitting criteria (Gini and Entropy). For every configuration, accuracy and confusion matrices were generated to assess how effectively the model distinguished between Alzheimer's (class 1) and non-Alzheimer's (class 0) cases.

The accuracy results for each configuration are summarized below:

Split Ratio	Criterion	Accuracy
90/10	Gini	0.9488
90/10	Entropy	0.9488
80/20	Gini	0.9465
80/20	Entropy	0.9348
70/30	Gini	0.9488
70/30	Entropy	0.9410

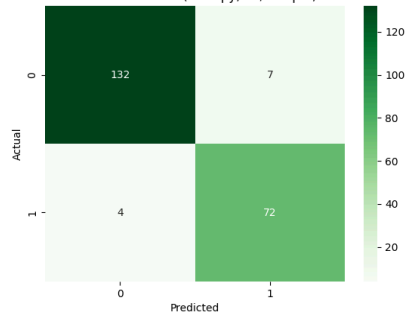
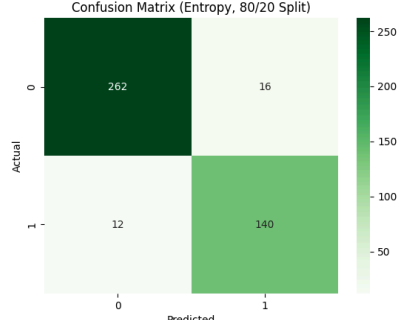
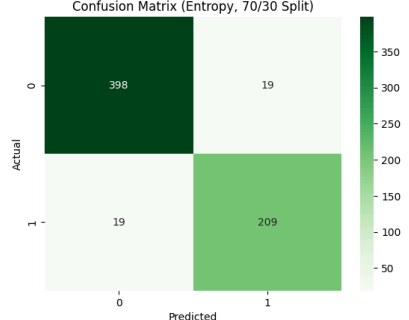
1.Gini Index

Split Ratio	Confusion Matrix	Accuracy	Explanation
90/10		%94.88 (0.9488)	Using the Gini criterion with a 90/10 split achieved a high accuracy of 94.88%. The confusion matrix shows very few misclassifications, indicating that having more training data helped the model learn strong and reliable patterns.
80/20		%94.65 (0.9465)	With the 80/20 split, accuracy slightly dropped to 94.65%. The confusion matrix shows a small rise in errors, suggesting that reducing the training size slightly affects the model's stability, though performance remains strong.
70/30		%94.88 (0.9488)	The 70/30 split also reached 94.88% accuracy, matching the 90/10 result. The confusion matrix shows consistent predictions across both classes, indicating that the model generalizes well even with a larger test set.

Best Gini Result

The best Gini results were obtained with both the 90/10 and 70/30 splits, each achieving an accuracy of **0.9488**. Since both configurations performed equally well, the Gini criterion can be considered stable and effective across different data partitions.

2.Information Gain Criterion (Entropy)

Split Ratio	Confusion Matrix	Accuracy	Explanation									
90/10	Confusion Matrix (Entropy, 90/10 Split)  <table><tr><th>Actual \ Predicted</th><th>0</th><th>1</th></tr><tr><th>0</th><td>132</td><td>7</td></tr><tr><th>1</th><td>4</td><td>72</td></tr></table>	Actual \ Predicted	0	1	0	132	7	1	4	72	%94.88 (0.9488)	With the Entropy criterion and a 90/10 split, the model achieved an accuracy of 0.9488, matching the best Gini results. The confusion matrix shows minimal errors, indicating that Entropy performs reliably when trained on a larger portion of the dataset.
Actual \ Predicted	0	1										
0	132	7										
1	4	72										
80/20	Confusion Matrix (Entropy, 80/20 Split)  <table><tr><th>Actual \ Predicted</th><th>0</th><th>1</th></tr><tr><th>0</th><td>262</td><td>16</td></tr><tr><th>1</th><td>12</td><td>140</td></tr></table>	Actual \ Predicted	0	1	0	262	16	1	12	140	%93.48 (0.9348)	Using an 80/20 split with Entropy resulted in a lower accuracy of 0.9348, with the confusion matrix showing more misclassified samples than the 90/10 split. This indicates that Entropy is more sensitive to training size and performs better with larger training sets.
Actual \ Predicted	0	1										
0	262	16										
1	12	140										
70/30	Confusion Matrix (Entropy, 70/30 Split)  <table><tr><th>Actual \ Predicted</th><th>0</th><th>1</th></tr><tr><th>0</th><td>398</td><td>19</td></tr><tr><th>1</th><td>19</td><td>209</td></tr></table>	Actual \ Predicted	0	1	0	398	19	1	19	209	%94.11 (0.9411)	The 70/30 split using Entropy achieved an accuracy of 0.9411, performing better than the 80/20 split but still below the 90/10 result. The confusion matrix shows acceptable performance with slightly more errors, indicating that Entropy is reasonably effective but less consistent than Gini across different splits.
Actual \ Predicted	0	1										
0	398	19										
1	19	209										

Best Entropy Result

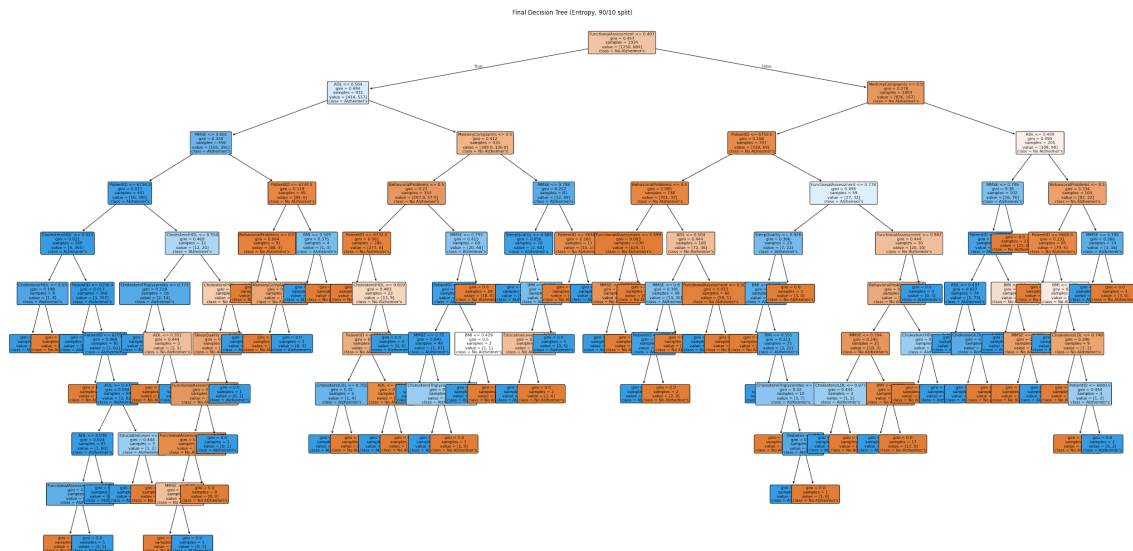
The best Entropy configuration was the 90/10 split, achieving the highest accuracy of 0.9488. This confirms that the Entropy-based model benefits significantly from a larger training portion.

Best Overall Decision Tree Configuration (Gini vs. Entropy)

Across all experiments, both **Gini** and **Entropy** produced strong and very similar results; however, a closer examination of the accuracy scores and confusion matrices reveals that the **Gini** criterion demonstrated slightly more consistent performance across different data splits. Gini reached its highest accuracy of **0.9488** with both the **90/10** and **70/30** configurations, and in each case the confusion matrices showed consistently low misclassification rates, indicating stable and reliable predictions.

Entropy also achieved a high accuracy of **0.9488** with the **90/10** split, but its performance declined more noticeably in the **80/20** and **70/30** splits. This suggests that **Entropy** is more sensitive to changes in training size, while **Gini** maintains strong results even when the training proportion varies.

Overall, the best-performing and most stable configuration is **Gini** with the **90/10** split, as it combines the highest accuracy with consistently balanced predictions. **Based on this evaluation, the final Decision Tree model can be confidently selected using the Gini criterion and a 90/10 train–test split for optimal performance on this dataset.**



B. Clustering Results (K-Means)

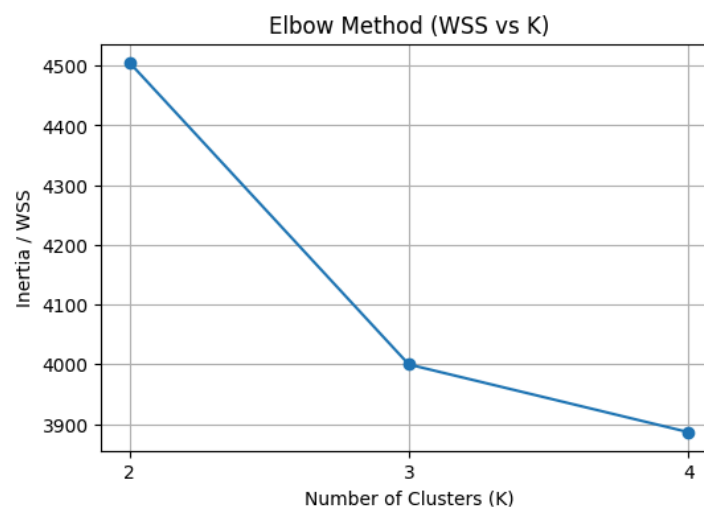
The K-Means algorithm was applied to explore natural groupings of patients based on their medical and demographic characteristics. To determine the most suitable number of clusters, three values of K (2, 3, and 4) were tested. Each configuration was evaluated using:

- Inertia (WSS): Measures how compact the clusters are
- Silhouette Score: Measures how well-separated the clusters are

The results are summarized in the table below:

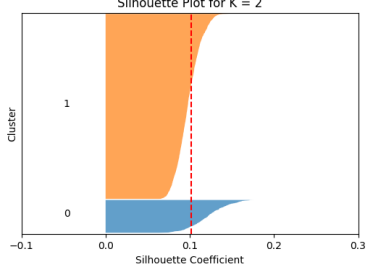
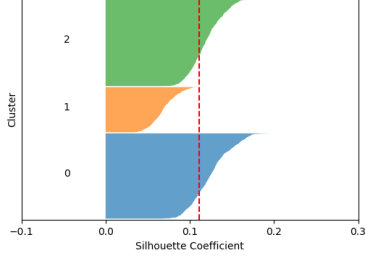
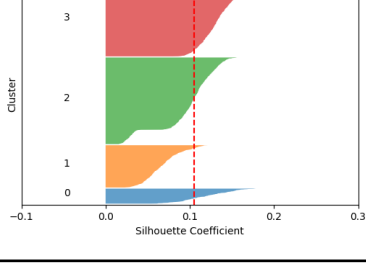
K	Inertia	Silhouette Score
2	4505.35	0.1076
3	3999.83	0.1155
4	3886.55	0.1051

• 1. Elbow Method



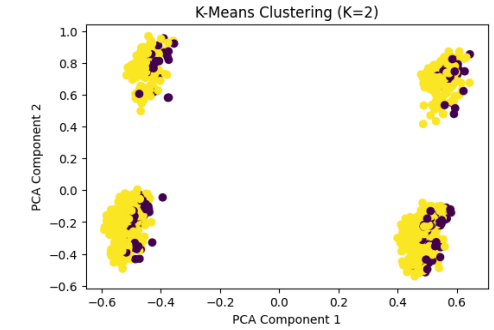
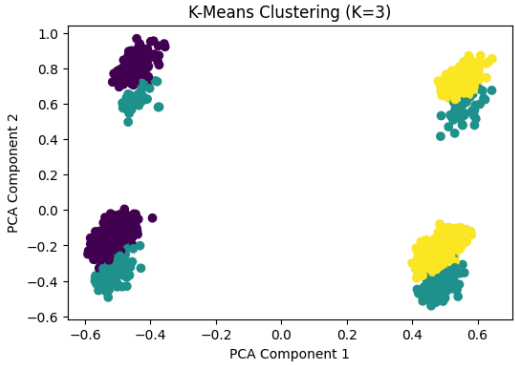
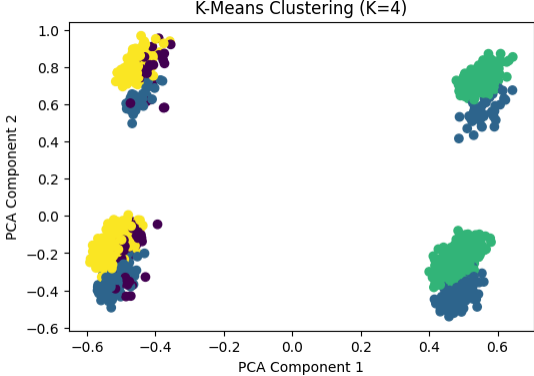
The Elbow plot shows that the decrease in WSS slows noticeably after K = 3, indicating diminishing improvement when additional clusters are added. This suggests that K = 3 provides a balanced and meaningful clustering structure.

• 2.Silhouette

Value of K	Visualization	Average Silhouette width
K = 2		0.1076
K = 3		0.1155
K = 4		0.1051

The silhouette plots for K = 2, 3, and 4 show that K = 3 provides slightly better cluster separation than K = 2 and K = 4. Although all average silhouette scores are relatively low, K = 3 has the highest average value and more compact clusters, which supports selecting K = 3 as the optimal number of clusters for this dataset.

• 3.K-Means clustering

Value of K	Cluster Visualization	Explanation
k = 2	 K-Means Clustering (K=2) The scatter plot shows two clusters of data points (yellow and purple) in a 2D space defined by PCA Component 1 and PCA Component 2. The clusters are broad and overlap significantly, indicating weak separation.	For K = 2, the Silhouette score is low (0.1076) and the Inertia is 4505.35. The visualization shows two broad clusters with noticeable overlap, indicating weak separation. This means the model does not clearly distinguish patient profiles when only two clusters are used.
k = 3	 K-Means Clustering (K=3) The scatter plot shows three distinct clusters of data points (yellow, purple, and teal) in a 2D space defined by PCA Component 1 and PCA Component 2. The clusters are well-separated and compact, suggesting that K = 3 provides the most meaningful and distinguishable patient clusters.	For K = 3, the Silhouette score rises to 0.1155—the highest among all tested values—and the Inertia drops to 3999.83, indicating better cluster compactness. The visualization shows three well-separated groups, suggesting that K = 3 provides the most meaningful and distinguishable patient clusters.
k = 4	 K-Means Clustering (K=4) The scatter plot shows four clusters of data points (yellow, purple, green, and blue) in a 2D space defined by PCA Component 1 and PCA Component 2. The clusters are positioned close together, indicating more overlap and weaker separation than with K = 3.	For K = 4, the Silhouette score drops to 0.1051, even though the Inertia decreases further to 3886.55. The visualization shows clusters positioned close together, indicating more overlap and weaker separation than with K = 3. This suggests that K = 4 does not provide as meaningful clustering.

Selecting the Optimal Number of Clusters (Majority Rule)

To determine the best value of K, we evaluated all three tested options (K = 2, 3, and 4) using three criteria:

- Silhouette Score (higher is better) — Best result: K = 3
- Inertia / WSS (lower is better) — Best result: K = 4
- Visual inspection of PCA plots — Best separation: K = 3

These results show mixed outcomes, so a majority-rule approach was used.

Since two out of three criteria (Silhouette and visualization) favored K = 3, it was selected as the optimal number of clusters for this dataset.

Best Overall K-Means Configuration

After comparing K = 2, 3, and 4, the most effective clustering configuration is K = 3.

This conclusion is supported by:

- Highest Silhouette Score (0.1155) Indicates the best-defined cluster structure
- Acceptable Inertia value : While K = 4 had lower Inertia, clusters were less separated
- Clear visual separation : K = 3 produced clusters that were more distinct and meaningful, with minimal overlap

K = 2 generated clusters that were too broad, while K = 4 resulted in clusters that were too close together and harder to interpret.

In contrast, K = 3 provided the best balance between compactness and separation.

Therefore, K = 3 is selected as the optimal and most interpretable clustering solution for this dataset.

A sample of the dataset with the assigned cluster labels (K = 3) is shown below.

Gender	EducationLevel	BMI	SleepQuality	FamilyHistoryAlzheimers	CardiovascularDisease	Diabetes	HeadInjury	Hypertension
3.108624e-15	0.433888	0.499826	0.504641	0.248219	0.128266	0.162708	0.093824	0.100000
5.100671e-01	0.428784	0.522708	0.498670	0.228188	0.165548	0.138702	0.080537	0.100000
1.000000e+00	0.424031	0.503859	0.516917	0.268605	0.148837	0.145349	0.097674	0.100000

7. Findings

A. Classification Findings (Decision Tree)

The Decision Tree classifier was tested using three different train–test splits (90/10, 80/20, and 70/30) and two splitting criteria (Gini and Entropy). Across all experiments, the Gini criterion with the 90/10 split consistently produced the highest and most stable accuracy (0.9488). The confusion matrices for each configuration showed very low levels of misclassification, confirming that the model was able to distinguish Alzheimer’s cases from non-Alzheimer’s cases with high reliability.

Analysis of the final tree structure highlighted that several cognitive and functional features were the most influential in determining the diagnosis. The top predictors included MMSE score, ADL score, Functional Assessment, Memory Complaints, and Behavioral Problems. These findings align with clinical understanding, where cognitive decline and impaired daily functioning are central indicators of Alzheimer’s Disease.

Key insight:

The Decision Tree model demonstrated strong predictive performance and provided clear, interpretable decision rules, making it highly suitable for Alzheimer’s prediction.

B. Clustering Findings (K-Means)

K-Means clustering was evaluated using $K = 2, 3$, and 4 , and the performance of each configuration was assessed using Silhouette Scores, Inertia values, Silhouette plots, and PCA visualizations. Among the tested values, $K = 3$ was identified as the most effective and meaningful grouping.

- $K = 2$ resulted in clusters with substantial overlap.
- $K = 3$ achieved the highest Silhouette Score (0.1155) and produced clearer visual separation.
- $K = 4$ had lower Silhouette performance and more overlapping boundaries despite lower Inertia.

The cluster centers indicated three distinct patient profiles:

1. A group with generally lower risk levels and better lifestyle factors (e.g., higher sleep quality).
2. A group with higher medical risks, including cardiovascular conditions and diabetes.
3. A group with characteristics more strongly associated with Alzheimer’s, such as family history and lower cognitive scores.

Key insight: $K = 3$ provides the most meaningful clustering structure and reflects natural differences in patients’ health and cognitive profiles.

C. Best-Performing Models

- **Best Classification Model:**
Decision Tree (Gini, 90/10) — delivered the highest accuracy, the fewest misclassifications, and the most interpretable model structure.
- **Best Clustering Configuration:**
K-Means with $K = 3$ — produced the best Silhouette Score, balanced Inertia, and the clearest separation between groups.

Together, these models provide both strong predictive performance and valuable insights into the underlying patterns in the dataset.

D. Relation to the Selected Research Paper

The findings of this project align closely with the selected research paper [1]. The research emphasizes the importance of medical, cognitive, and lifestyle attributes in predicting Alzheimer's risk—consistent with the influential features identified in our Decision Tree model, such as sleep quality, age, family history, cardiovascular conditions, and cognitive scores.

Additionally, the paper highlights the need for interpretable and transparent models in clinical settings. This directly supports the use of a Decision Tree, which is known for its clear structure and ease of interpretation. Although the research paper focuses primarily on classification, it acknowledges the diversity among Alzheimer's patients—a point reinforced by our clustering results, where $K = 3$ revealed three meaningful patient groups with distinct characteristics.

E. Overall Interpretation

When combined, the classification and clustering results provide a comprehensive view of Alzheimer's patterns within the dataset. The Decision Tree offers a reliable and interpretable predictive model, while the K-Means clusters help highlight natural groupings and variations among patient profiles. These findings are consistent with existing literature and emphasize the importance of analyzing both medical and lifestyle factors when assessing Alzheimer's risk.

Overall, the models used in this project offer meaningful and credible insights that support early detection and contribute to a deeper understanding of Alzheimer's Disease.

8. Conclusion

In conclusion, this project allowed us to apply the full data mining process that we learned throughout the course and achieve the goals we aimed for at the beginning. We worked step by step through the main stages of data mining, and each stage contributed to building a clear understanding of the dataset and the patterns related to Alzheimer's Disease.

Throughout the project, we completed the following main steps:

- **Data Understanding:**
Explored the dataset, examined feature types, and identified initial patterns.
- **Data Preprocessing:**
Handled missing values and outliers, discretized and normalized features, and performed encoding and feature selection.
- **Classification (Decision Tree):**
Built and evaluated a predictive model that achieved strong and stable performance.
- **Clustering (K-Means):**
Explored natural patient groupings and selected the most meaningful number of clusters.

By completing these steps, we were able to extract useful insights from the data and apply the techniques covered in the course in a practical and structured way. Overall, the project helped us connect theory to real-world data and provided a clear final result that meets the objectives of the assignment.

9. References

- [1] Y. Dubey, A. Bhongade, P. Palsodkar, and P. Fulzele, "Efficient Explainable Models for Alzheimer's Disease Classification with Feature Selection and Data Balancing Approach Using Ensemble Learning," *Diagnostics*, vol. 14, no. 2770, pp. 1–15, Dec. 2024. doi: 10.3390/diagnostics14242770.
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